

CSE 4512: Computer Networks Lab

Lab 02 Report

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1 Lab Task

The objective of this lab is to design a network topology for "Iutians Assemble," a company expanding its headquarters across five floors. The design requires Variable Length Subnet Masking (VLSM) to efficiently allocate IP addresses based on the specific device requirements of each department.

Given the Student ID **18**, the base network prefix is calculated as:

$$192.(18).(18 + 3 \pmod{14}) \rightarrow \mathbf{192.18.7.0}$$

The device requirements are calculated based on the provided formulas:

- **Engineering (Floor 1):** $E(18) = (7(18) + 3) \pmod{41} + 10 = 16$ devices.
- **Marketing (Floor 2):** $M(18) = (5(18) + 11) \pmod{26} + 5 = 28$ devices.
- **Sales (Floor 3):** $S(18) = (11(18) + 7) \pmod{36} + 5 = 30$ devices.
- **Administration (Floor 4):** $A(18) = (3(18) + 17) \pmod{18} + 3 = 20$ devices.
- **Management (Floor 5):** $G(18) = (13(18) + 1) \pmod{9} + 2 = 3$ devices.

2 Network Topology

Below is the screenshot of the final network topology designed in Cisco Packet Tracer.

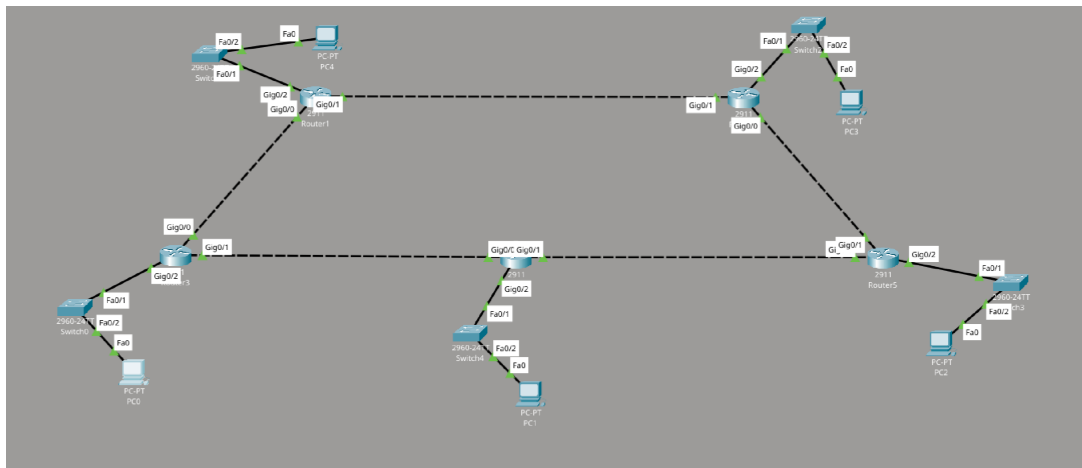


Figure 1: Final Network Topology for Iutians Assemble

3 Procedure

3.1 Step 1: VLSM Calculation

We sorted the departments by the number of required host addresses (descending order) to ensure the most efficient use of the IP space. The base network is 192.18.7.0.

1. **Sales (30 hosts):** Requires 32 IPs (Block size). Mask: /27.

2. **Marketing (28 hosts):** Requires 32 IPs. Mask: /27.
3. **Administration (20 hosts):** Requires 32 IPs. Mask: /27.
4. **Engineering (16 hosts):** Requires 32 IPs. Mask: /27.
5. **Management (3 hosts):** Requires 8 IPs. Mask: /29.

3.2 Step 2: Subnetting Worksheet Configuration

Using the calculations above, the subnet table was populated:

Department	Dev	Network	Mask	Usable Host Range
Sales	30	192.18.7.0	/27	192.18.7.1 – 192.18.7.30
Marketing	28	192.18.7.32	/27	192.18.7.33 – 192.18.7.62
Administration	20	192.18.7.64	/27	192.18.7.65 – 192.18.7.94
Engineering	16	192.18.7.96	/27	192.18.7.97 – 192.18.7.126
Management	3	192.18.7.128	/29	192.18.7.129 – 192.18.7.134

Table 1: Completed Subnetting Worksheet

3.3 Step 3: Topology Setup

1. Launched Cisco Packet Tracer.
2. Placed 1 Router (Generic) to act as the central gateway.
3. Placed 5 Switches, one for each department.
4. Connected at least one PC to each switch to represent the end devices.
5. Connected switches to the Router GigabitEthernet interfaces.

3.4 Step 4: Device Configuration

1. **Router Configuration:** Assigned the first usable IP of each subnet to the respective router interface (Default Gateway).
 - Sales Gateway: 192.18.7.1
 - Marketing Gateway: 192.18.7.33
 - Admin Gateway: 192.18.7.65
 - Engineering Gateway: 192.18.7.97
 - Management Gateway: 192.18.7.129
2. **PC Configuration:** Assigned a usable IP to the PCs and set the Default Gateway to the router IP configured above. Subnet masks were set according to the /27 or /29 CIDR values.

4 Observations & Results

To verify the network flow, ping tests were initiated from the **Sales Department PC** to the erst of the departments.

```
C:\>ping 192.18.7.40

Pinging 192.18.7.40 with 32 bytes of data:

Reply from 192.18.7.40: bytes=32 time<1ms TTL=126
Reply from 192.18.7.40: bytes=32 time<1ms TTL=126
Reply from 192.18.7.40: bytes=32 time<1ms TTL=126
Reply from 192.18.7.40: bytes=32 time<1ms TTL=126

Ping statistics for 192.18.7.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 2: Successful Ping from Sales (192.18.7.10) to Marketing (192.18.7.40)

```
C:\>ping 192.18.7.70

Pinging 192.18.7.70 with 32 bytes of data:

Reply from 192.18.7.70: bytes=32 time<1ms TTL=125
Reply from 192.18.7.70: bytes=32 time<1ms TTL=125
Reply from 192.18.7.70: bytes=32 time<1ms TTL=125
Reply from 192.18.7.70: bytes=32 time<1ms TTL=125

Ping statistics for 192.18.7.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 3: Successful Ping from Sales (192.18.7.10) to Administration (192.18.7.70)

```
C:\>ping 192.18.7.130

Pinging 192.18.7.130 with 32 bytes of data:

Reply from 192.18.7.130: bytes=32 time=12ms TTL=126
Reply from 192.18.7.130: bytes=32 time=1ms TTL=126
Reply from 192.18.7.130: bytes=32 time<1ms TTL=126
Reply from 192.18.7.130: bytes=32 time=1ms TTL=126

Ping statistics for 192.18.7.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Figure 4: Successful Ping from Sales (192.18.7.10) to Engineering (192.18.7.100)

```
C:\>ping 192.18.7.100

Pinging 192.18.7.100 with 32 bytes of data:

Reply from 192.18.7.100: bytes=32 time<1ms TTL=125
Reply from 192.18.7.100: bytes=32 time<1ms TTL=125
Reply from 192.18.7.100: bytes=32 time<1ms TTL=125
Reply from 192.18.7.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.18.7.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 5: Successful Ping from Sales (192.18.7.10) to Management (192.18.7.130)

As seen in the screenshot, the packets were successfully transmitted and received, confirming that the routing table is correctly populated and the gateways are functioning.

5 Challenges Faced

- **Router RIP Setup:** I set the router to router interface ip addresses in 192.168.0.0 network, which I needed to specify in the cli when using router rip command.